

Decent Coaching Classes

X Semi

Tuesday Holiday Homework

Part I: Ch 1 (Linear Equations in Two Variables) & Ch 2 (Quadratic Equations)

Chapter 1 — Linear Equations in Two Variables

1. Solve by the graphical method: $x + y = 6$, $x - y = 2$.
2. Solve by the elimination method: $4x + 3y = 18$, $2x - 3y = 0$.
3. Solve by Cramer's Rule: $3x + y = 11$, $x + 2y = 7$.
4. Solve by the substitution method: $2x + y = 7$, $x - y = 2$.
5. For what value of k will the pair of equations $2x + 3y = 7$ and $kx + 6y = 14$ have infinitely many solutions?
6. The cost of 3 pencils and 2 pens is ₹36, while the cost of 5 pencils and 4 pens is ₹66. Find the cost of one pencil and one pen (form and solve the equations).
7. A two-digit number is 4 more than 6 times the sum of its digits. If the digits are interchanged, the resulting number is 18 less than the original number. Find the original number.

Chapter 2 — Quadratic Equations

8. Solve by factorisation: $x^2 - 9x + 20 = 0$
9. Solve using the formula: $3x^2 - 5x - 2 = 0$
10. Find the value of the discriminant for the equation $2x^2 + 5x + 3 = 0$, and state the nature of its roots.
11. If one root of the quadratic equation $x^2 - 8x + k = 0$ is 3, find the value of k and the other root.
12. The product of two consecutive positive even integers is 168. Form a quadratic equation and find the integers.
13. Find two numbers whose sum is 27 and product is 182 (form and solve a quadratic equation).
14. The sum of the squares of two consecutive natural numbers is 313. Find the numbers.

ANSWER KEY

Chapter 1 — Linear Equations in Two Variables

1. Adding the two equations: $2x = 8 \Rightarrow x = 4$; substituting: $y = 2$. Solution: $x = 4, y = 2$ (the two lines intersect at $(4, 2)$).
2. Adding the two equations: $6x = 18 \Rightarrow x = 3$; substituting in $2x - 3y = 0$: $6 - 3y = 0 \Rightarrow y = 2$. Solution: $x = 3, y = 2$.
3. $D = (3)(2) - (1)(1) = 5$; $D_x = (11)(2) - (1)(7) = 15$; $D_y = (3)(7) - (11)(1) = 10$. $x = D_x/D = 3, y = D_y/D = 2$.
4. From eqn 2: $x = y + 2$. Substitute in eqn 1: $2(y+2) + y = 7 \Rightarrow 2y + 4 + y = 7 \Rightarrow 3y = 3 \Rightarrow y = 1, x = 3$.
5. For infinitely many solutions, ratios of coefficients must be equal: $2/k = 3/6 = 7/14$. From $3/6 = 7/14$ (both equal $1/2$), and $2/k = 1/2 \Rightarrow k = 4$.
6. Let cost of 1 pencil = x , 1 pen = y . $3x + 2y = 36$ and $5x + 4y = 66$. Multiply first eqn by 2: $6x + 4y = 72$. Subtracting second: $x = 6$. Substituting: $18 + 2y = 36 \Rightarrow y = 9$. Pencil = ₹6, Pen = ₹9.
7. Let the number be $10x + y$. Condition 1: $10x + y = 6(x+y) + 4 \Rightarrow 10x + y = 6x + 6y + 4 \Rightarrow 4x - 5y = 4$. Condition 2: $10x + y - (10y + x) = 18 \Rightarrow 9x - 9y = 18 \Rightarrow x - y = 2 \Rightarrow x = y + 2$. Substituting: $4(y+2) - 5y = 4 \Rightarrow 4y + 8 - 5y = 4 \Rightarrow -y = -4 \Rightarrow y = 4, x = 6$. The number is 64.

Chapter 2 — Quadratic Equations

8. $x^2 - 9x + 20 = 0 \Rightarrow (x-4)(x-5) = 0 \Rightarrow x = 4$ or $x = 5$.
9. $a=3, b=-5, c=-2$. $x = [5 \pm \sqrt{(25+24)}]/6 = [5 \pm 7]/6 \Rightarrow x = 2$ or $x = -1/3$.
10. $D = b^2 - 4ac = 25 - 24 = 1$. Since $D > 0$ and is a perfect square, the roots are real, unequal and rational.
11. Substituting $x=3$: $9 - 24 + k = 0 \Rightarrow k = 15$. Equation: $x^2 - 8x + 15 = 0 \Rightarrow (x-3)(x-5) = 0$. The other root is 5.
12. Let integers be x and $x+2$: $x(x+2) = 168 \Rightarrow x^2 + 2x - 168 = 0 \Rightarrow (x-12)(x+14) = 0 \Rightarrow x = 12$ (rejecting negative). Integers: 12 and 14.
13. Let numbers be x and $27-x$: $x(27-x) = 182 \Rightarrow 27x - x^2 = 182 \Rightarrow x^2 - 27x + 182 = 0 \Rightarrow (x-13)(x-14) = 0 \Rightarrow x = 13$ or 14 . Numbers: 13 and 14.
14. Let numbers be x and $x+1$: $x^2 + (x+1)^2 = 313 \Rightarrow x^2 + x^2 + 2x + 1 = 313 \Rightarrow 2x^2 + 2x - 312 = 0 \Rightarrow x^2 + x - 156 = 0 \Rightarrow (x-12)(x+13) = 0 \Rightarrow x = 12$ (rejecting negative). Numbers: 12 and 13.